SHEAVES OF ABELIAN GROUPS ON A SITE HAVE FUNCTORIAL INJECTIVE EMBEDDINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction of the functorial embedding; the injectivity check is carried out through the criterion proved earlier in the chapter, carried on this card as a navigation stub.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The criterion for injectivity of a sheaf. Same chapter, Lemma lemma-characterize-injectives. Supplies the test for injectivity on subsheaves of the standard objects that the construction verifies.
- The sheaves associated to standard objects. Same chapter, the abelian sheaves associated to representable presheaves and the discussion of their subsheaves. Supplies the family of sheaves out of which the candidate injective sheaf is assembled.
- Products of injective sheaves. Same chapter, the product rule for injective objects. Supplies that a product of injective sheaves is injective, which gives injectivity of the constructed sheaf.

Source reference. Original definition 0139: sources/stacks-homology/pinned-original-source.tex, source lines 7355–7355. Navigation pointer invoked at proof line 1063; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01DN: sources/stacks-injectives/pinned-original-source.tex, source lines 957–957. Navigation pointer invoked at proof line 1070; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 01DO: sources/stacks-injectives/pinned-original-source.tex, source lines 1018–1018. Navigation pointer invoked at proof line 1080; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01DM: sources/stacks-injectives/pinned-original-source.tex, source lines 927–927. Navigation pointer invoked at proof line 1085; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Theorem 0.1. *The category of sheaves of abelian groups on a site has enough injectives. In fact there exists a functorial injective embedding, see Homology, Definition 0139.*

Proof. Let $\mathcal{G}_i$, $i \in I$ be a set of abelian sheaves such that every subsheaf of every $\mathbf{Z}_X^\#$ occurs as one of the $\mathcal{G}_i$. Apply Lemma 01DN to this collection to get an ordinal β. We claim that for any sheaf of abelian groups $\mathcal{F}$ the map $\mathcal{F} \to J_\beta(\mathcal{F})$ is an injection of $\mathcal{F}$ into an injective. Note that by construction the assignment $\mathcal{F} \mapsto \big(\mathcal{F} \to J_\beta(\mathcal{F})\big)$ is indeed functorial.

The proof of the claim comes from the fact that by Lemma 01DO it suffices to extend any morphism $\gamma : \mathcal{G} \to J_\beta(\mathcal{F})$ from a subsheaf $\mathcal{G}$ of some $\mathbf{Z}_X^\#$ to all of $\mathbf{Z}_X^\#$. Then by Lemma 01DN the map γ lifts into $J_\alpha(\mathcal{F})$ for some $\alpha < \beta$. Finally, we apply Lemma 01DM to get the desired extension of γ to a morphism into $J_{\alpha+1}(\mathcal{F}) \to J_\beta(\mathcal{F})$. $\square$